

PACKAGES

1.Create Package PK1 with below 3 Procedures.

1.Procedures ADD_DEPT : This procedures will take 3 IN arguments DNO,NAME,LO and duly insert these values into the DEPT table. Handle the exception if the user tries to insert a duplicate row using User Defined Exception Handler.

2.Procedure UPDATE_DEPT which takes 3 arguments DNO,NAME,LO. Upadte the DNAME,LOC for the corresponding DEPTNO.Handle exception when user enters DEPTNO that does not exists.

3.Procedures DELETE_DEPT which takes 1 argument DNO.Delete the record for the corresponding DEPTNO.Handle the required exception when user enters a not existing DEPTNO.

4.Create a Local Function CAL_CNT that returns True if DEPTNO does not Exists ELSE return FALSE.

Use this function all 3 Procedures.Create a One Time procedure that display the total number of rows.

```
SQL> CREATE OR REPLACE PACKAGE PK1 AS
2
3     PROCEDURE ADD_DEPT
4     (
5         DNO IN DEPT.DEPTNOMTYPE,
6         NAME IN DEPT.DNAME%TYPE,
7         LO IN DEPT.LOC%TYPE
8     );
9
10    PROCEDURE UPDATE_DEPT
11    (
12        DNO IN DEPT.DEPTNOMTYPE,
13        NAME IN DEPT.DNAME%TYPE,
14        LO IN DEPT.LOC%TYPE
15    );
16
17    PROCEDURE DELETE_DEPT
18    (
19        DNO IN DEPT.DEPTNOMTYPE
20    );
21
22 END PK1;
23 /
Package created.

SQL> SET SERVEROUTPUT ON;
SQL>
SQL> CREATE OR REPLACE PACKAGE BODY PK1 AS
2
3     -- Local Function
4     FUNCTION CAL_CNT(DNO NUMBER) RETURN BOOLEAN IS
5         V_CNT NUMBER;
6     BEGIN
7         SELECT COUNT(*)
8         INTO V_CNT
9         FROM DEPT
```

DATE:28-07-2026

```
11      IF V_CNT = 0 THEN
12          RETURN TRUE;      -- Department does not exist
13      ELSE
14          RETURN FALSE;    -- Department exists
15      END IF;
16  END CAL_CNT;
17
18  -----
19  -- ADD_DEPT
20  -----
21
22  PROCEDURE ADD_DEPT
23  (
24      DNO IN DEPT.DEPTNO%TYPE,
25      NAME IN DEPT.DNAME%TYPE,
26      LO IN DEPT.LOC%TYPE
27  )
28  IS
29      DUP_DEPT EXCEPTION;
30  BEGIN
31
32      IF NOT CAL_CNT(DNO) THEN
33          RAISE DUP_DEPT;
34      END IF;
35
36      INSERT INTO DEPT
37      VALUES(DNO, NAME, LO);
38
39      COMMIT;
40
41      DBMS_OUTPUT.PUT_LINE('Department inserted successfully.');
```

```
48  -----
49  -- UPDATE_DEPT
50  -----
51
52  PROCEDURE UPDATE_DEPT
53  (
54      DNO IN DEPT.DEPTNO%TYPE,
55      NAME IN DEPT.DNAME%TYPE,
56      LO IN DEPT.LOC%TYPE
57  )
58  IS
59      DEPT_NOT_FOUND EXCEPTION;
60  BEGIN
61
62      IF CAL_CNT(DNO) THEN
63          RAISE DEPT_NOT_FOUND;
64      END IF;
65
66      UPDATE DEPT
67      SET DNAME = NAME,
68          LOC = LO
69      WHERE DEPTNO = DNO;
70
71      COMMIT;
72
73      DBMS_OUTPUT.PUT_LINE('Department updated successfully.');
```

DATE:28-07-2026

```
Command Prompt - sqlplus / > X + - >
86      IS
87      DEPT_NOT_FOUND EXCEPTION;
88      BEGIN
89
90          IF CAL_CNT(DNO) THEN
91              RAISE DEPT_NOT_FOUND;
92          END IF;
93
94          DELETE FROM DEPT
95          WHERE DEPTNO = DNO;
96
97          COMMIT;
98
99          DBMS_OUTPUT.PUT_LINE('Department deleted successfully.');
```

Package body created.

```
SQL> BEGIN
```

```
Command Prompt - sqlplus / > X + - >
111      SELECT COUNT(*)
112      INTO V_TOTAL
113      FROM DEPT;
114
115      DBMS_OUTPUT.PUT_LINE('Total Rows in DEPT = ' || V_TOTAL);
116  END;
117
118  END PK1;
119  /
```

Package body created.

```
SQL> BEGIN
2      PK1.ADD_DEPT(50,'HR','HYDERABAD');
3  END;
4  /
```

Total Rows in DEPT = 4
Department inserted successfully.
PL/SQL procedure successfully completed.

```
SQL> BEGIN
2      PK1.UPDATE_DEPT(50,'HRM','BANGALORE');
3  END;
4  /
```

Department updated successfully.
PL/SQL procedure successfully completed.

```
SQL> BEGIN
2      PK1.DELETE_DEPT(50);
3  END;
4  /
```

Department deleted successfully.
PL/SQL procedure successfully completed.

```
SQL> |
```

DATE:28-07-2026

2.Create Package PK2 that has 2 Procedures DISP_FIRST3 and DISP_FIRST6 that will display the first 3 employees and first 6 employees from EMP Table.Create a Global Cursor.

```
SQL> CREATE OR REPLACE PACKAGE PK2 AS
2
3     CURSOR C_EMP IS
4         SELECT EMPNO, ENAME, JOB, SAL
5         FROM EMP
6         ORDER BY EMPNO;
7
8     PROCEDURE DISP_FIRST3;
9     PROCEDURE DISP_FIRST6;
10
11 END PK2;
12 /

Package created.

SQL> CREATE OR REPLACE PACKAGE BODY PK2 AS
2
3     PROCEDURE DISP_FIRST3 IS
4         V_EMPNO EMP.EMPNO%TYPE;
5         V_ENAME EMP.ENAME%TYPE;
6         V_JOB EMP.JOB%TYPE;
7         V_SAL EMP.SAL%TYPE;
8         CNT NUMBER := 0;
9     BEGIN
10         OPEN C_EMP;
11
12         LOOP
13             FETCH C_EMP INTO V_EMPNO, V_ENAME, V_JOB, V_SAL;
14             EXIT WHEN C_EMP%NOTFOUND OR CNT = 3;
15
16             DBMS_OUTPUT.PUT_LINE(
17                 V_EMPNO || ' ' || V_ENAME || ' ' ||
18                 V_JOB || ' ' || V_SAL
19             );
20
21             CNT := CNT + 1;
22         END LOOP;
```

```
23
24     CLOSE C_EMP;
25     END DISP_FIRST3;
26
27     PROCEDURE DISP_FIRST6 IS
28         V_EMPNO EMP.EMPNO%TYPE;
29         V_ENAME EMP.ENAME%TYPE;
30         V_JOB EMP.JOB%TYPE;
31         V_SAL EMP.SAL%TYPE;
32         CNT NUMBER := 0;
33     BEGIN
34         OPEN C_EMP;
35
36         LOOP
37             FETCH C_EMP INTO V_EMPNO, V_ENAME, V_JOB, V_SAL;
38             EXIT WHEN C_EMP%NOTFOUND OR CNT = 6;
39
40             DBMS_OUTPUT.PUT_LINE(
41                 V_EMPNO || ' ' || V_ENAME || ' ' ||
42                 V_JOB || ' ' || V_SAL
43             );
44
45             CNT := CNT + 1;
46         END LOOP;
47     CLOSE C_EMP;
48     END DISP_FIRST6;
49
50 END PK2;
51 /

Package body created.

SQL> SET SERVEROUTPUT ON;
SQL>
SQL> EXEC PK2.DISP_FIRST3;
7359 SMITH 800
```

DATE:28-07-2026

```
49
50 DBMS_OUTPUT.PUT_LINE(
51   V_EMPNO || ' ' || V_ENAME || ' ' ||
52   V_JOB || ' ' || V_SAL
53 );
54
55 CNT := CNT + 1;
56 END LOOP;
57
58 CLOSE C_EMP;
59 END DISP_FIRST6;
60
61 END PK2;
62 /
Package body created.

SQL> SET SERVEROUTPUT ON;
SQL>
SQL> EXEC PK2.DISP_FIRST6;
7359 SMITH 800
7499 ALLEN SALESMAN 1600
7521 WARD SALESMAN 1250

PL/SQL procedure successfully completed.

SQL>
SQL> EXEC PK2.DISP_FIRST6;
7359 SMITH 800
7499 ALLEN SALESMAN 1600
7521 WARD SALESMAN 1250
7566 JONES MANAGER 3421.25
7664 MARTIN SALESMAN 1250
7698 BLAKE MANAGER 2850

PL/SQL procedure successfully completed.

SQL> |
```

3.Create a sequence S2 start with 1 and increment by 1.Create a package PK3 that has 2 procedures with the same name ADD_EMP. The procedure will take 3 arguments EMPNO,ENAME DEFAULT 'UNK',SAL DEFAULT 1000. This procedure inserts these values in the EMP table.The second procedure accepts 2 arguments NAME DEAFULT 'UNK',SAL DEFAULT 1000 and inserts into the EMP table calling the sequence S2 for EMPNO column.Also insert other ENAME AND SAL columns.

```
SQL> CREATE SEQUENCE S2
2 START WITH 1
3 INCREMENT BY 1;
Sequence created.

SQL> CREATE OR REPLACE PACKAGE PK3 AS
2
3   PROCEDURE ADD_EMP(
4     P_EMPNO EMP.EMPNO%TYPE,
5     P_ENAME EMP.ENAME%TYPE DEFAULT 'UNK',
6     P_SAL EMP.SAL%TYPE DEFAULT 1000
7   );
8
9   PROCEDURE ADD_EMP(
10    P_ENAME EMP.ENAME%TYPE DEFAULT 'UNK',
11    P_SAL EMP.SAL%TYPE DEFAULT 1000
12  );
13
14 END PK3;
15 /
Package created.

SQL> CREATE OR REPLACE PACKAGE BODY PK3 AS
2
3   -- Procedure 1
4   PROCEDURE ADD_EMP(
5     P_EMPNO EMP.EMPNO%TYPE,
6     P_ENAME EMP.ENAME%TYPE DEFAULT 'UNK',
7     P_SAL EMP.SAL%TYPE DEFAULT 1000
8   )
9   IS
10  BEGIN
11    INSERT INTO EMP (EMPNO, ENAME, SAL)
12    VALUES (P_EMPNO, P_ENAME, P_SAL);
13
```

DATE:28-07-2026

```
12     VALUES (P_EMPNO, P_ENAME, P_SAL);
13
14     DBMS_OUTPUT.PUT_LINE('Employee inserted successfully.');
```

```
15 END ADD_EMP;
16
17
18 -- Procedure 2 (Uses Sequence S2)
19 PROCEDURE ADD_EMP2
20     P_ENAME EMP.ENAME%TYPE DEFAULT 'UNK',
21     P_SAL    EMP.SAL%TYPE DEFAULT 1000
22 )
23 IS
24 BEGIN
25     INSERT INTO EMP (EMPNO, ENAME, SAL)
26     VALUES (S2.NEXTVAL, P_ENAME, P_SAL);
27
28     DBMS_OUTPUT.PUT_LINE('Employee inserted using Sequence S2.');
```

```
29 END ADD_EMP2;
30
31 END PK3;
32 /
```

Package body created.

```
SQL> SET SERVEROUTPUT ON;
SQL>
SQL> EXEC PK3.ADD_EMP(9001, 'RAHUL', 2500);
Employee inserted successfully.

PL/SQL procedure successfully completed.

SQL> EXEC PK3.ADD_EMP('ANITA', 3000);
Employee inserted using Sequence S2.

PL/SQL procedure successfully completed.

SQL> EXEC PK3.ADD_EMP(9002);
Employee inserted successfully.
```

```
PL/SQL procedure successfully completed.

SQL> EXEC PK3.ADD_EMP(9002);
Employee inserted successfully.

PL/SQL procedure successfully completed.

SQL>
SQL> EXEC PK3.ADD_EMP;
Employee inserted using Sequence S2.

PL/SQL procedure successfully completed.

SQL> SELECT EMPNO, ENAME, SAL
2 FROM EMP
3 ORDER BY EMPNO;
```

EMPNO	ENAME	SAL
1	ANITA	3000
2	UNK	1000
7359	SMITH	800
7499	ALLEN	1600
7521	MARO	1250
7566	JONES	3021.25
7654	MARTIN	1250
7698	BLAKE	2850
7788	SCOTT	3450
7789	VAISH	57500
7844	TURNER	1500

EMPNO	ENAME	SAL
7982	FORD	3450
9001	RAHUL	2500
9002	UNK	1000

14 rows selected.